

Be the Agent

Play an AI agent with cards and find out what the model does and what the tools do

An AI agent such as Claude Code gets a goal instead of a recipe. It plans its own steps, uses tools and keeps going until the job is done. In this experiment you play such an agent yourselves, with cards, a calculator and a few printed pages instead of a computer. One of you is the language model, one is everything around it, and one watches and keeps the log. By the end you will have run, by hand, the loop that every AI agent runs on.

Goal

Find out from the inside how an agent works: what the language model decides, what the tools do, and why everything the model knows has to be on its stack of cards.

What you need

- A group of three. In a group of four, two of you share the Toolbox.
- The kit at the end of this document, printed and cut out: the system card, three task cards, the farm files, the web pages, the rules for the Toolbox and the Observer's sheet.
- A stack of blank index cards, a pen for everyone and a calculator.
- About 30 minutes for the first two tasks.

The three roles

Role	What you do	What you may look at
Model	You decide what happens next and write it on a card.	Only the stack of cards in front of you.
Toolbox	You carry out exactly what a tool card says and write the result on a new card.	The farm files, the web pages and the calculator.
Observer	You keep the log: every card in order, with a note on what happened.	Everything, but you never help.

The rules

1. **The Model sees only its stack.** It never looks at the farm files or the web pages, and nobody talks to it. Everything it learns arrives on a card.
2. **Every card the Model writes starts with one of three words:**
 - **THINK:** a note to itself, for example **THINK: I need the size of the field first.**
 - **TOOL:** a tool call, written exactly as on the system card, for example **TOOL: read_file("fields").**
 - **ANSWER:** the answer to the task. This card ends the game.
3. **The Toolbox carries out a tool card literally** and writes the result on a new card that starts with **RESULT:.** If the tool does not exist or the call is written wrongly, the result is an error. The Toolbox never repairs a call, never guesses what the Model meant and never adds anything that was not asked for.
4. **Every card goes on top of the stack** in the order it was written, and it stays there.
5. **One card per turn.** The Model writes one card; if it is a tool card, the Toolbox answers it before the Model writes the next one.

Steps

1. **Set up.** The Model's stack starts with the system card, with the card for Task 1 on top of it. The Toolbox lays out the farm files and the web pages where the Model cannot see them. The Observer takes the log sheet.

2. **Play Task 1.** How many cards did it take? Did the Model write a **THINK:** card before its first tool call?
3. **Swap roles and play Task 2.** Which tools did the Model use, and in which order? Did the first search find the right page? Which value did the Model choose, and how did it decide?
4. **Count.** The Observer reads the log aloud: rounds, tool calls, errors, and how many cards lie on the stack at the end.
5. **Play Task 3 if there is time.** One thing in the farm files will only be noticed by a careful Model.
6. **Compare with a real agent.** Your lecturer shows the log of a real agent session. Find the places where it thinks, calls a tool, receives a result and answers.

What you just played

In the game	In a real AI agent
the stack of cards	the context window : everything the model knows during a task
the system card	the system prompt : who the model is and which tools it has
the task card	the user's prompt , the goal
a THINK: card	the model's reasoning or plan
a TOOL: card	a tool call : the model only writes text, a program carries it out
the Toolbox	the harness , the program around the model that runs the tools
a RESULT: card	the observation : the tool's result, added to the context
the Observer's log	the transcript of the session
the ANSWER: card	the stop condition : the goal is reached

The loop you ran, think, act and observe until the goal is reached, is the loop inside every AI agent. The Model never opened a file and never pressed a key on the calculator; it only wrote text, and something else carried it out.

Go further

1. **Limited memory.** The stack may now hold at most eight cards. When it is full, the Model writes a **SUMMARY:** card that replaces the oldest cards; the system card and the task card always stay. Play Task 3 again. What got lost, and did it matter?
2. **Be the tool for a chatbot.** Solve Task 2 with a chatbot in your browser. Whenever it needs a file, a web page or a calculation, you have to provide it yourself. Who is the Toolbox now, and who decides the next step?
3. **A new tool.** Invent a sixth tool, for example `forecast("date")`, add its description to the system card and write a page for the Toolbox. Does the Model use the new tool without being told to?
4. **A wrong result.** The Toolbox secretly returns one wrong result, for example a calculation that is off by a factor of ten. Does the Model notice? What would a real agent need in order to notice?

Reflection

Write down your own answers first, then discuss them in your group.

1. In which round would the Model have been stuck without the Toolbox's cards?
2. What would have gone differently if the Model had had to give up its stack after ten cards?
3. Who did the thinking, who did the acting, and which of it could the Model actually do?

The kit

Print this part and cut out the cards. All farm data and all values on the web pages are invented for this game. They are simplified teaching values and not advice for real fertilisation.

System card

SYSTEM

You are the assistant of Greenfield Farm. You can only act by writing cards. On every turn, write exactly one card that starts with **THINK:**, **TOOL:** or **ANSWER:**.

You can use these tools:

- `search("words")` finds pages on the web and returns their titles
- `open("title")` returns the full text of one web page
- `read_file("name")` returns one of the farm's files: `fields`, `soil_tests`, `stock`, `weather`
- `calculator("expression")` calculates, for example `calculator("4.5 * 100")`
- `write_file("name", "text")` saves a text as a new file of the farm

When you have answered the task, write **ANSWER:** and stop.

Task cards

TASK 1

How large is the field Long Acre, and what grows on it this year?

TASK 2

How much nitrogen may we still apply to Long Acre this spring? Give the total in kg N for the whole field.

TASK 3

Plan the first nitrogen application on Long Acre: how many kilograms of which fertiliser product do we need, and can we spread it on 10 February? Save the plan as the file `plan`.

Rules for the Toolbox

Carry out each tool card exactly as written and answer with a card that starts with **RESULT:**.

Tool	What you do	If it goes wrong
<code>search("words")</code>	Compare the search words with the keyword line of every web page. Write down the titles of all pages that contain at least one of the words, one title per line.	RESULT: no pages found
<code>open("title")</code>	Copy the whole text of the page with this title.	RESULT: error, page not found
<code>read_file("name")</code>	Copy the whole file.	RESULT: error, file not found
<code>calculator("expression")</code>	Calculate exactly what is written and round to two decimals.	RESULT: error, cannot calculate
<code>write_file("name", "text")</code>	Write the text on a new sheet with the name as its title and put it with the farm files.	–

A call with a tool that is not in this list, or written differently, gets **RESULT:** error, unknown tool.

Farm files (Toolbox only)

fields		
Field	Size	Crop 2026
Long Acre	4.5 ha	winter wheat
Mill Field	3.2 ha	oilseed rape
Back Meadow	2.0 ha	grassland

soil_tests		
Field	Sampled	Soil mineral nitrogen (Nmin)
Long Acre	3 February 2026	40 kg N/ha
Mill Field	3 February 2026	25 kg N/ha

stock		
Product	Nitrogen content	In store
Calcium ammonium nitrate (CAN)	27 %	2,000 kg
Urea	46 %	500 kg

weather

Date	Weather
9 February	frost at night, ground frozen until midday
10 February	frost day and night, ground frozen all day
11 February	mild, ground thawed

Web pages (Toolbox only)

Nitrogen requirement of arable crops — teaching guide 2026

Keywords: nitrogen, requirement, crops, wheat, barley, rape, maize, beet, fertiliser

The nitrogen requirement is the amount of nitrogen a crop needs in one season, in kg N per hectare. From it, subtract the soil mineral nitrogen (N_{min}) measured in spring; the rest may be applied as fertiliser.

Crop	Requirement
winter wheat	180 kg N/ha
winter barley	160 kg N/ha
oilseed rape	200 kg N/ha
grain maize	170 kg N/ha
sugar beet	150 kg N/ha

Fertiliser recommendations 2009 (archive)

Keywords: nitrogen, fertiliser, wheat, barley, recommendation, archive

This page is no longer updated. Recommended nitrogen for winter wheat: 210 kg N/ha; for winter barley: 190 kg N/ha.

Splitting nitrogen in winter wheat

Keywords: wheat, nitrogen, split, dose, timing, first

Winter wheat gets its nitrogen in three doses: the first dose, 40 % of the total, at the start of growth in spring; the second, 40 %, when the stem starts to grow; the third, 20 %, when the ears appear.

When may fertiliser be spread? — teaching rules

Keywords: when, date, rules, ban, spreading, frozen, snow, fertiliser

No nitrogen fertiliser may be spread between 1 October and 31 January. At any time of year, fertiliser must never be spread on ground that is frozen, flooded or covered in snow.

Calcium ammonium nitrate (CAN)

Keywords: CAN, calcium, ammonium, nitrate, product, content, fertiliser

Calcium ammonium nitrate contains 27 % nitrogen: 100 kg of product bring 27 kg of nitrogen to the field.

Winter wheat varieties 2026

Keywords: wheat, varieties, yield, 2026

Twelve new varieties were tested this year. The highest yields came from varieties with good winter hardiness and resistance to yellow rust.

Observer’s sheet

Task: _____ Model: _____ Toolbox: _____

Round	Card	What it said, in short	Note: error, detour, good move
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			

Rounds: _____ Tool calls: _____ Errors: _____ Cards on the stack at the end: _____